

Lecture- 22

Mathematics 1 (15B11MA111)

**Module: Vector Differential
and Integral Calculus**

CO [C105.4]

Topics to be covered

➤ Surface Integrals

- ❖ Evaluation of Surface Integrals

- ❖ Examples

- ❖ Practice Problems

➤ Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Surface Integral

Surface integral of a vector function $\vec{F}$ over the surface S is given as the integral of the component of $\vec{F}$ along the normal to the surface and is written as

$$\iint_S \vec{F} \cdot \hat{n} \, dS$$

Where $\hat{n}$ is unit normal vector to S and

$$dS = \frac{dx \, dy}{\hat{n} \cdot \hat{k}} \text{ (projection on xy-plane)}$$

$$dS = \frac{dy \, dz}{\hat{n} \cdot \hat{i}} \text{ (projection on yz-plane)}$$

$$dS = \frac{dx \, dz}{\hat{n} \cdot \hat{j}} \text{ (projection on xz-plane)}$$

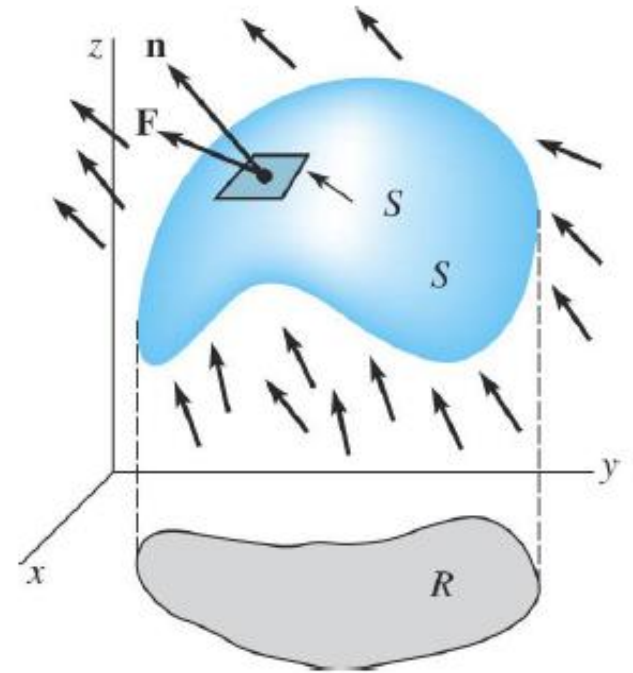


Image: <https://www.math.upenn.edu/~wziller/math114f13>

And double integral is performed over projection R .

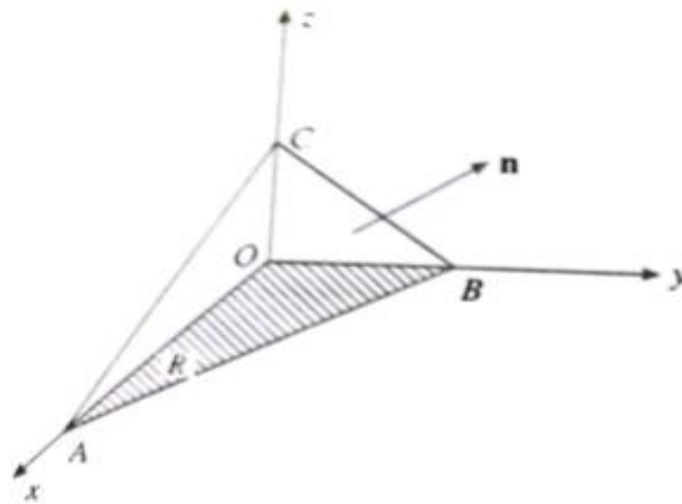
The total volume of the fluid flowing through the surface S per unit time is called the flux of $\vec{F}$ through S and it is given by above surface integral.

Example 1: Evaluate the surface integral

$$\iint_S F \cdot n \, dA$$

Where $F = 6zi + 6j + 3yk$ and S is the portion of the plane $2x + 3y + 4z = 12$, which is in the first octant.

Solution: Let $f(x,y,z) = 2x + 3y + 4z - 12 = 0$ be the surface. Then



$$\text{grad } f = 2i + 3j + 4k, n = \frac{\text{grad } f}{|\text{grad } f|} = \frac{1}{\sqrt{29}}(2i + 3j + 4k)$$

The projection of the plane ABC in the first octant is the triangle bounded by $x=0$, $y=0$ and $2x+3y=12$. We have ,

$$dA = \frac{dx dy}{n.k} = \frac{dx dy}{4/\sqrt{29}}$$

so,
$$\iint_S F \cdot n \, dA = \iint_S \frac{1}{\sqrt{29}} (12z + 18 + 12y) dA$$

From the surface equation, we have $4z = 12 - 2x - 3y$.

$$\iint_S F \cdot n \, dA = \iint_S \frac{1}{\sqrt{29}} (54 - 6x + 3y) dA$$

$$= \frac{1}{4} \iint_S (54 - 6x + 3y) dx \, dy$$

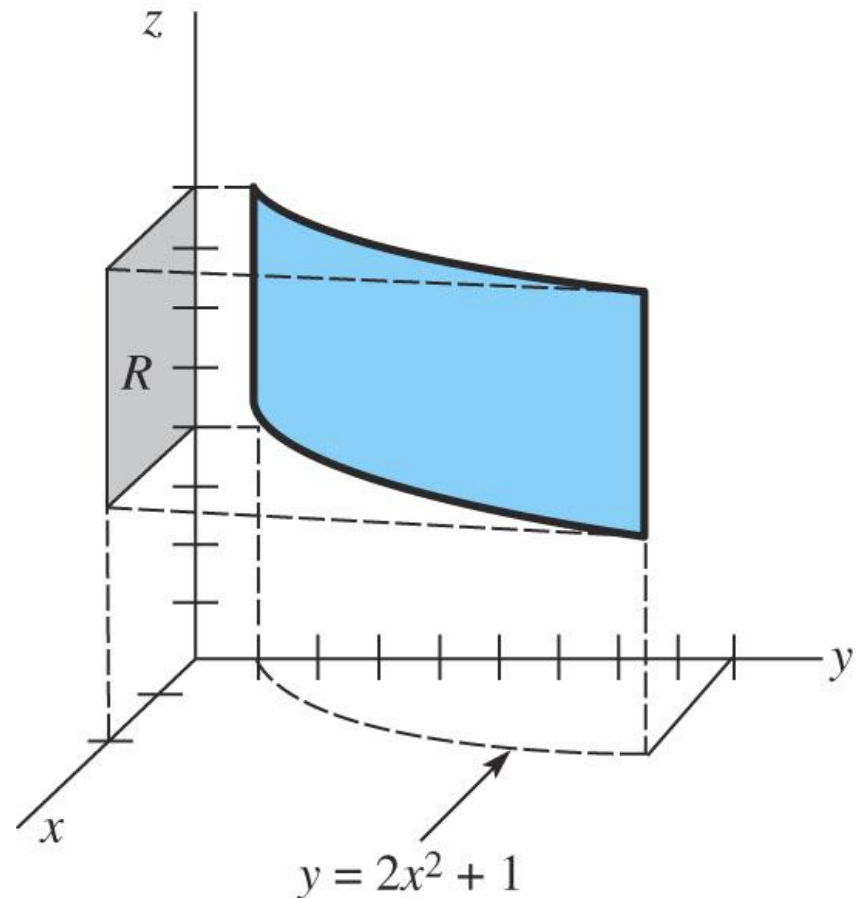
$$\iint_S F \cdot n \, dA = \frac{1}{4} \int_{x=0}^6 \left[\int_{y=0}^{(12-2x)/3} (54 - 6x + 3y) dy \right] dx = 138$$

Example 2 surface integral

Evaluate $\iint_S xz^2 dS$, where S is the portion of $y = 2x^2 + 1$ in the first octant bounded by $x = 0$, $x = 2$, $z = 4$ and $z = 8$.

Solution

The projection graph on the xz -plane is shown in Fig 1.



Example (2)

Let $f(x, y, z) = 2x^2 + 1 - y$. Since $f_x(x, y, z) = 4x$, $f_y(x, y, z) = -1$ and $f_z(x, y, z) = 0$, then

$$\begin{aligned} \text{grad } f &= 4xi - j + 0k, n = \frac{\text{grad } f}{|\text{grad } f|} \\ &= \frac{1}{\sqrt{1 + 16x^2}} (4xi - j + 0k) \\ ds &= \frac{dx dz}{n \cdot j} = \sqrt{1 + 16x^2} dx dz \end{aligned}$$

$$\iint_S xz^2 \, dS = \int_0^2 \int_4^8 xz^2 \sqrt{1 + 16x^2} \, dz \, dx$$

$$= \int_0^2 \left[\frac{z^3}{3} x \sqrt{1 + 16x^2} \right]_4^8 dx = \frac{448}{3} \int_0^2 x(1 + 16x^2)^{1/2} dx$$

$$= \frac{28}{9} (1 + 16x^2)^{3/2} \Big|_0^2 = \frac{28}{9} [65^{3/2} - 1] \approx 1627.3$$

Practice Problems

Q.1: Evaluate the surface integral

$$\iint_S \vec{F} \cdot \vec{dS}$$

Where $\vec{F} = 18zi - 12j + 3yk$ and S is the portion of the plane $2x+3y+6z=12$, which is in the first octant.

Q.2: Evaluate the surface integral

$$\iint_S \vec{F} \cdot \vec{n} \, dA$$

Where $\vec{F} = z^2\vec{i} + xy\vec{j} - y^2\vec{k}$ and S is the portion of the surface of the cylinder $x^2 + y^2 = 36$, $0 \leq z \leq 4$ included in the first octant.

Answers: 1) 24

2) 416

Thank You